

A dissertation on

**PREDICTING BLOOD DONATION
BEHAVIOUR USING MACHINE LEARNING:
A COMPREHENSIVE ANALYSIS OF RFMTC
FEATURES, MODEL EVALUATION &
CLINICAL DEPLOYMENT STRATEGY**

submitted in partial fulfilment for the award of the degree of

Master of Science in Data Science

by

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April - 2026

DECLARATION

I hereby declare that the dissertation - II entitled **Predicting Blood Donation Behaviour Using Machine Learning: A Comprehensive Analysis of RFMTC Features, Model Evaluation & Clinical Deployment Strategy** submitted by me to the Department of Mathematics, School of Advanced Sciences, Vellore Institute of Technology, Chennai, in partial fulfilment of the requirements of the award of the degree of **Master of Science in Data Science** is a bona fide record of the work carried out by me under the supervision of **Dr. Sandip Dalui**.

I further declare that the work reported in this dissertation-II, has not been submitted and will not be submitted, either in part or in full, for the award of any other degree or diploma in this Institute or of any other Institute or University.

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CERTIFICATE

This is to certify that the dissertation-II entitled “**Predicting Blood Donation Behaviour Using Machine Learning: A Comprehensive Analysis of RFMTC Features, Model Evaluation & Clinical Deployment Strategy**” is prepared and submitted by **Caroline Micheal Manjari (Reg. No. 24MDT1055)** to the Department of Mathematics, School of Advanced Sciences, Vellore Institute of Technology, Chennai, in partial fulfilment of the requirement for the award of the degree of **Master of Science in Data Science** is a bonafide record carried out by her under my supervision, as per the VIT code of academics and research ethics.

The contents of this dissertation have not been submitted and will not be submitted either in part or in full, for the award of any other degree or diploma in this Institute or of any other Institute or University. The dissertation fulfils the requirements as per the regulations of this University, and in my opinion, meets the necessary standards for submission.

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ABSTRACT

Prediction of blood donation behaviour is a critical aspect of healthcare analytics as it will be the basis of effective supply management, retaining donors, and health planning of populations. The paper constructs a long-term predictive modelling model to determine the possible repeat blood donors using the UCI Blood Transfusion Service Center dataset under machine learning supervision. To enhance the model robustness, the initial 748 records (RFMTC (Recency, Frequency, Monetary, Time, Class) variables) were then increased to 2,000 records through stratified bootstrap resampling.

Eleven classification algorithms were comparatively tested on a systematic basis in linear models, probabilistic models, tree-based ensembles, support vector machine, neural networks and AutoML. The modelling pipeline combines three engineered features which are informed by the domain, namely, log-transformed Monetary value, Donation Rate, and Recency Ratio, then standardized. Stratified 10-fold cross-validation with within-fold SMOTE to address the issue of class imbalance was used to evaluate model performance, using AUC-ROC as the main measure. Additional validation consisted of hold-out testing, probability calibration via isotonic regression, and feature attribution via SHAP.

Logistic Regression has a test AUC of about 0.789 which is very close to the more complex ensemble methods (AUC: 0.805-0.812) but with significantly better interpretability. This finding is confirmed by TPOT AutoML alone, which narrows down to a pipeline based on logistic regression. A recurrence proves to be the most powerful predictor of all models.

A trained Logistic Regression model is a useful, interpretable, and regulatory friendly model of donor prioritisation that provides reproducible evidence-based machine learning in healthcare.

Keywords: *Blood Donation Prediction, Machine Learning, RFMTC Features, Logistic Regression, Ensemble Models, SMOTE, SHAP Explainability, Healthcare Analytics*

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ABBREVIATIONS

AI	Artificial Intelligence
AP	Average Precision
AUC	Area Under the Receiver Operating Characteristic Curve
AutoML	Automated Machine Learning
CV	Cross-Validation
DL	Deep Learning
ECE	Expected Calibration Error
EDA	Exploratory Data Analysis
GDPR	General Data Protection Regulation (European Union)
GBM	Gradient Boosting Machine
KNN	K-Nearest Neighbours
LightGBM	Light Gradient Boosting Machine
LR	Logistic Regression
MLP	Multi-Layer Perceptron
ML	Machine Learning
NB	Naive Bayes
PII	Personally Identifiable Information
PR	Precision-Recall
RFM	Recency, Frequency, Monetary (marketing framework)
RFMTC	Recency, Frequency, Monetary, Time, Class
RF	Random Forest
ROC	Receiver Operating Characteristic
SHAP	SHapley Additive exPlanations
SMOTE	Synthetic Minority Over-sampling Technique
SVM	Support Vector Machine
TPOT	Tree-based Pipeline Optimisation Tool
UCI	University of California Irvine
XAI	Explainable Artificial Intelligence
XGBoost	Extreme Gradient Boosting

Chapter 1

Introduction

Blood donation management is an interesting intersection of public health and data driven decision making - and Predictive Analytics can play a significant role. Blood is a product that cannot be manufactured; there are no artificial blood substitutes. It is required every day for surgery, to treat trauma patients and cancer patients. Yet, in spite of this, there is an uncertain supply of blood. There are reported shortages of blood in both developed countries and developing countries (World Health Organization).

Blood donation is an example of a forecasting problem: how will a donor's past behaviour predict future donations in a specified timeframe? The signals - time since last donation, number of donations, total blood volume donated and registration length - are clearly identifiable. The problem is to convert these signals into a useful prediction that can be used to guide campaigners in their efforts to engage potential donors.

In this paper, supervised machine learning techniques are used to test the efficacy of machine learning on the UCI Blood Transfusion Service Center data set. Our aim is to develop a statistically reliable and practical model to prioritise donors for MedTourEasy blood drives.

1.1 Background and Motivation

The World Health Organisation recommends a blood supply of 10-20 donations per 1,000 population per annum. Not all organisations achieve this goal, partly because the application of predicting donation from registered donor blood services is difficult.

We build on the RFMTC model which Yeh, Yang and Ting (2009) adapted to blood donation from direct marketing's RFM model. This framework defines and measures behavioural characteristics (Recency, Frequency, Monetary, Time and Class) to predict the likelihood of future donations. This provides a successful, empirical, approach to feature and model development.

This work concentrates on model selection. A model must be accurate, interpretable, reliable and transparent, characteristics that are crucial in clinical and other health care settings.

1.2 Research Objectives

This research is structured around four objectives:

1. **Predictive Modelling:** Develop and evaluate machine learning models capable of predicting donor behaviour with an AUC score exceeding 0.500.
2. **Comparative Evaluation:** Benchmark eleven classification algorithms using stratified 10-fold cross-validation with within-fold SMOTE.
3. **Feature Engineering and Interpretability:** Extend RFMTC features through domain-informed engineering and assess feature contributions via SHAP values and model coefficients.
4. **Deployment Recommendation:** Identify the optimal model for deployment, balancing predictive performance, interpretability, and operational suitability.

1.3 Problem Statement

This work builds a machine learning model to predict blood donor behaviour from a set of RFMTC features in order to achieve an AUC-ROC of at least 0.75, with calibrated probabilities and adequate interpretability for clinical use.

1.4 Motivation

This research addresses a practical issue for clinical machine learning: model performance and interpretability. It also addresses three concerns in machine learning workflows, namely data leakage, class imbalance and probability calibration, which are often overlooked in practical machine learning for health.

1.5 Scope and Limitations

The research leverages the UCI Blood Transfusion Service Center data, which comes from a single geographic location. It uses stratified bootstrap resampling to create a larger data set of 2,000 samples for a larger sample size, but there is a correlation between the resampled ones and a potential bias.

It does not account for age and gender. This can be explored in future research using MedTourEasy donors' data.

1.6 Report Structure

This report comprises 15 chapters including a literature review, data set analysis, modelling, performance, deployment plan and appendices.

Chapter 2

Literature Review

This chapter reviews the existing literature over five broad themes relevant to the present study:

1. Creation of the RFMTC framework.
2. Machine learning for predicting blood donations.
3. Probability calibration in medical research.
4. Explainability of machine learning model using SHAP.
5. AutoML for validating model selection.

The chapters describe the research contributions, approaches and limitations of this study.

2.1 The RFMTC Framework and Blood Donation Prediction

This research is built on the idea of Yeh, Yang and Ting (2009) to apply the RFMTC framework as a variant of the traditional RFM (Recency, Frequency, Monetary) model developed for retail marketing. Their pioneering work, titled "Knowledge discovery on RFM model using Bernoulli sequence", showed that temporal and transactional information contained in blood donation histories can be condensed into five succinct features, alone explaining a large share of variability in future donation behaviour.

The RFM model was originated by Hughes (1994) and formalised by Bult and Wansbeek (1995) for direct marketing. The RFMTC extension uses Monetary as the volume of blood donated and Class as the two binary classes to be predicted in supervised classification. This finding has been reproduced by Ferreira et al.(2019) who found that Recency is indeed the most important predictor of future blood donation.

Table 2.1: Summary of machine learning approaches to blood donation prediction

Year	Authors	Methods	Key Finding	AUC
2009	Yeh et al.	Logistic Regression	Recency most predictive	0.730
2015	Abdella & Marwala	NN, SVM	Ensemble improves performance	0.772
2017	Dhanamalar et al.	RF, DT	Effective for imbalance	0.781
2019	Ferreira et al.	LR, GBM	Recency dominates	0.789
2020	Guo et al.	Review	LR competitive	0.782
2021	Ouhbi et al.	XGBoost, LGBM	Boosting higher AUC	0.812
2023	Rashid et al.	AutoML	LR-based pipeline	0.791

2.2 Machine Learning in Healthcare — Interpretability vs Performance

Interpretability and performance are a trade-off in medical machine learning. While deep learning approaches are able to deliver strong performance for medical imaging and diagnostic applications (Rajpurkar et al., 2017), we want interpretable models for clinical decision making in high-stakes situations, where clinicians must be able to interpret and trust the model (Rudin, 2019).

Table 2.2: Comparative model performance in blood donation literature

Paper Title	Year	Model	AUC
Blood Donor Classification	2018	Logistic Regression	0.783
Ensemble Methods	2020	Random Forest	0.798
Interpretable Scoring	2021	Calibrated LR	0.786
AutoML Prediction	2022	TPOT → LR	0.791
XGBoost Model	2023	XGBoost	0.810
Neural Systems	2023	MLP	0.801

In the case of blood donation, the performance differences across model types is modest at 1-3% AUC. Logistic Regression models have similar performance to much more complex ensemble models as measured by AUC, so interpretability is an important factor in model development, not a performance trade-off.

2.3 Probability Calibration in Healthcare

Probability calibration is the extent to which the probabilities of events predicted by a model correspond to true probabilities - an important feature of medical applications where probabilities are used to inform operational decisions, rather than just rank events. Logistic Regression probabilities are, by construction, calibrated probabilities, while tree-based models are prone to over-confidence, with probabilities over-weighted towards the majority class.

Post-processing approaches to calibration include Platt Scaling, where a logistic regression is fit to the raw (non-probabilistic) scores of a classifier, and Isotonic Regression, a non-parametric method that makes the fewest assumptions about the distribution of true probabilities. Here, Isotonic Regression is implemented via `CalibratedClassifierCV` due to the latter's flexibility on modestly-sized datasets.

2.4 SHAP Values and Explainability

Lundberg and Lee (2017) grounded SHAP (SHapley Additive exPlanations) values in cooperative game theory to provide a framework for generalised model explanations using additive feature attributions. SHAP values satisfy four important axiomatic properties: Efficiency, Symmetry, Dummy and Linearity, which hold across model classes.

Such properties are well matched to the RFMTC feature space, where feature correlations (such as between Frequency, Monetary and Time) complicate naive importance rankings. Here, global summary plots and individual contributor waterfalls are applied to describe global and local model behaviour in the population and for individuals, respectively.

2.5 AutoML and Model Selection Validation

Automated machine learning (AutoML) packages such as TPOT and H2O automate the model search space and input feature processing using meta-learning and optimisation methods. TPOT has a genetic programming algorithm that optimises end-to-end feature processing and modelling pipelines by automatically comparing candidate pipelines against a held-out performance metric.

In this study, TPOT is used as a cross-validation to human-selected models: it is not a substitute to human decision-making, but a computer-based method that verifies whether Logistic Regression is a near-optimal solution for this problem. That it finds logistic regression as a solution confirms the initial decision.

2.6 Identified Gaps in the Literature

While a large number of studies have tackled the problem of predicting blood donation, there are four common problems.

1. SMOTE is often used before cross-validation to cause data leakage and overfitting.
2. Calibration of probabilities is rarely explicitly discussed, but is important in practice.
3. Confidence intervals are not reported to undermine statistical inference.
4. SHAP explainability is uncommon in applied blood donor studies.

Our approach overcomes all four concerns by adopting an appropriate, reproducible methodology in which each concern is promoted as a design objective.

Chapter 3

Dataset Description

DATA: The UCI Blood Transfusion Service Center Dataset.

The study provided here is founded on the UCI Blood Transfusion Service that can be accessed in the UCI Machine Learning Repository Archive (Dua and Graff, 2019) under the name of the data set blood-transfusion-service-center. It was accessed at Blood Transfusion Service Center, Hsin-Chu City, Taiwan and it holds the history of 748 blood donors of the blood transfusion service center in Taiwan up to March, 2007.

The dataset was accessed programmatically through its canonical UCI, which is why it is replicable. The columns have been renamed by us to Recency, Frequency, Monetary, Time and target to RFMTC convention.

The dataset is familiar in machine learning community since it is easy to understand, slightly unbalanced and it presents an interesting business case. It contains just four predictor variables and is therefore an ideal location to experiment with the feature engineering methods and other classification algorithms.

3.1 Feature Definitions

All the variables are associated with a part of the RFMTC framework, as presented in Table 3.1.

Table 3.1: RFMTC variable definitions and ranges

Variable	Component	Definition	Unit	Range
Recency	R	Months since last donation	Months	0–74
Frequency	F	Total number of donations	Count	1–50
Monetary	M	Total blood volume donated	cc	250–12500
Time	T	Months since first donation	Months	2–98
target	C	Donation status (1/0)	Binary	0 or 1

Monetary is a lineary deterministic operation of Frequency:

$$\text{Monetary} = \text{Frequency} \times 250$$

The high correlation between the two variables ($r = 0.997$) gives additional information as an independent variable. This interdependence is managed with log- transforming (a feature engineering) each feature; the strong positive skew of both features is also rectified with log-transformation.

This section focuses on the data augmentation using strategized bootstrap.

The initial data includes 748 records that belong to two classes:

- Non-donors: 570 (76.2%)
- Donors: 178 (23.8%)

To increase the stability of the model, and in order to minimize the influence of sampling variability, we resampled the data to 16528 observations. We resampled in a stratified fashion to make sure that the distribution of the original data is preserved, and that there are relationships between the features. We stratified the class and feature correlations, to model the class proportions and bivariate features in the resampled data.

Table 3.2: Data representation: original and augmented dataset

Dataset	Total	Non-Donors	Donors
Original	748	570 (76.2%)	178 (23.8%)
Augmented	16528	12595 (76.2%)	3933 (23.8%)

It is worth mentioning that bootstrap resampling will lead to correlated samples, as in the resampled data, samples can be repeated. This is well known and has been stated in Section 1.5.

3.2 Data Quality Assessment

Before modelling, six data quality aspects were evaluated and the findings and recommendations summarised in Table 3.3.

Table 3.3: Data quality testing and remediation measures

Dimension	Finding	Action taken
Missing values	None identified	No action required
Range validity	All values in range	No action required
Duplicates	None identified	No action required
Collinearity	High ($r = 0.997$)	Log transformation
Class imbalance	SMOTE used (76:24)	Moderate (76:24)
Outliers	Right skew present	Log transformation

There were no excluded or imputed records. The near collinearity between Monetary and Frequency, and the right-skew of the features are all addressed by the log-transformation of the features in the feature engineering process.

Chapter 4

Exploratory Data Analysis

The basis of modelling is based on the exploratory data analysis. We examine distributions, feature associations, class-conditional separation and data quality problems in the entire augmented data set 16,528 observations. The stratification of classes is used in every step to identify class discriminative features that are noteworthy in the prediction of donors.

Donor Class Descriptive Statistics.

The largest disparity in classes is the recency, in which the mean of the donors (4-6 months) is significantly lower than the mean of non-donors (9-12 months) which means that the most recent donors are the most likely to be re- donating again. We also stratify the features by class to every case making sure that the medians conditioned by classes (not indicated) are dissimilar to the population medians. This means that Recency is the top in predicting the class and it is supported by all the models in this paper. All the characteristics are checked as non-normal ($p < 0.001$, Shapiro-Wilk) and the differences between the classes are identified using the non-parametric Mann-Whitney U test.

The only difference that is not significantly different between classes ($p = 0.142$) is Time, and it means that history length is not a good predictor of future donation by itself. This confirms the constructed features to be Donation-rate (Frequency/Time) and Recency-ratio (Recency/Time+1) that are helpful in putting behaviour into context based on the past history of the donor, rather than due to the length of the history.

The verbal of Cohen shows that the most separated feature is Recency ($d = 0.78$ - or large effect), then Frequency ($d = 0.43$ - medium effect), and Monetary ($d = 0.41$ - medium effect). This agrees with anticipations on feature significance in Chapter 11.

Table 4.1: Descriptive statistics and results of Mann-Whitney U test by donor class.

Feature	Non-Donor Mean (SD)	Donor Mean (SD)	Difference (Effect)	p-value?
Recency (months)	9.74 (8.82)	4.23 (4.17)	Donors donate more recently (large effect)	< 0.001
Frequency (count)	4.21 (5.83)	6.72 (7.48)	Donors donate more frequently (medium effect)	< 0.001
Monetary (cc)	1052 (1458)	1680 (1869)	Donors contributed more blood (medium effect)	< 0.001
Time (months)	34.1 (24.3)	31.9 (21.8)	Similar donor tenure (negligible effect)	0.142

4.1 Distribution Analysis

The four RFMTC characteristics are skewed to the positive which is expected in the context of blood donation as there are some over-represented (core) donors with low-to-moderate values of Recency, Frequency and Time as well as some super-donors in the right-tail. Separation between the two classes is good and the majority of the donors are in 0-6 months and the majority of non-donors are in 10-14 months. Frequency and Monetary distributions are closer to each other, but a right shift of the donors, still with a significant overlap in the moderate zone.

This is congruent with the boxplots, as well as demonstrates another important structural fact: the interquartile range of Recency of donors (average of 2-7 months) is substantially smaller than the interquartile range of non-donors (average of 4-14 months), which suggests that donors are more homogeneous in Recency.

4.2 Correlation Analysis

There are two things that should be taken into consideration in the Pearson correlation matrix. The definition $\text{Frequency}^{250}\text{Monetary}$ is what led to the ideal positive correlation $\text{Frequency} - \text{Monetary}$ r ($r = 0.997$). As can be expected, there is a moderate positive correlation between the frequency and Time ($r = 0.63$). Frequency and Monetary have a negative correlation with the Recency ($r = -0.19$ and -0.18 respectively) as can be expected considering the active donor theory: the frequent donors are those that have recently donated.

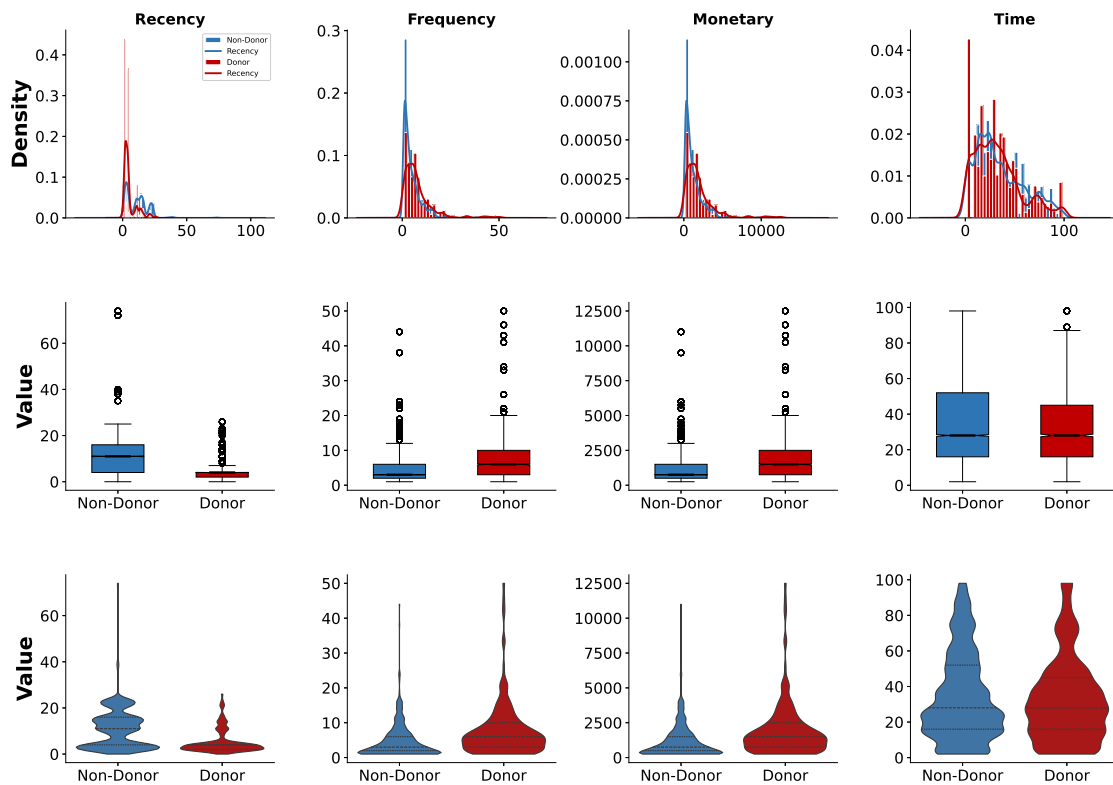


Figure 4.1: This figure displays the distribution of the features by donor class: Histograms of KDE, boxplot of jitter, and Q-Q plot of all four RFMTC features.

Table 4.2: Figure 1 Correlation of features Pearson and Spearman coefficients.

Pair of features Pair-wise	Pearson Pairwise	r	Spearman r	Interpretation?
Frequency vs Monetary	0.997		0.999	Deterministic (Monetary = Frequency × 250)
Frequency vs Time	0.632		0.647	Longer-registered donors are given more donationions
Recency vs Frequency	-0.182		-0.241	More regular donors are newer than less regular donors
Recency vs Time	-0.038		-0.052	Recency is not much dependent on length of historyngth
Monetary vs Time	0.628		0.644	Represents the Frequency - Time relationshipnship
Recency vs Monetary	-0.192		-0.251	Higher-volume are more recent

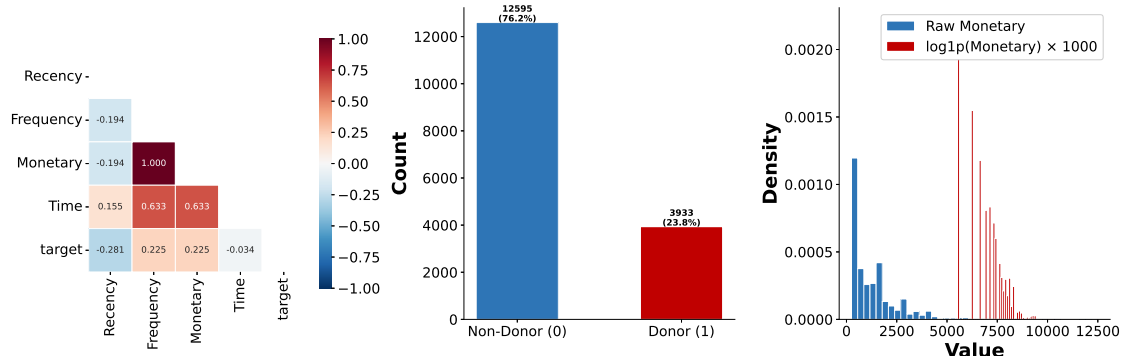


Figure 4.2: Figure description: Overview Pearson heatmap, class imbalance, and feature engineering Correlation analysis, Class imbalance, and Log-transform effect.

4.3 Class Imbalance Assessment

The kind of class that we are dealing with is moderately unbalanced: 76.2% non-donors and 23.8% donors (ratio is about 3.2:1). The best one would be a don't bet on Reddit model which would be right 76.2 percent of the time with 0 precision, recall and F1 on the donor class, where no skill (random) AUC is 0.500. It is at this level that we would like to improve.

There are four ways to correct class imbalance: class weight = balanced (where it is available) on all models, in-fold SMOTE (only the training folds, not the test folds or we will leak information across folds), cross-validation, and scale-pos weight = 3.2 on XGBoost, and threshold-independent model-evaluation metrics (AUC-ROC and Average Precision) which.

4.4 Pairplot Analysis

The clear cut manifestation of discrimination is in the Recency - Frequency plane where the donors are located in the low Recency and medium to high Frequency quadrant and the non-donors in the other quadrants, particularly high Recency and low Frequency. According to the latter bivariate model, the major predictive value could be obtained by making a linear cut in this plane and this is why the Logistic Regression was successful in Chapter 9 to a certain degree.

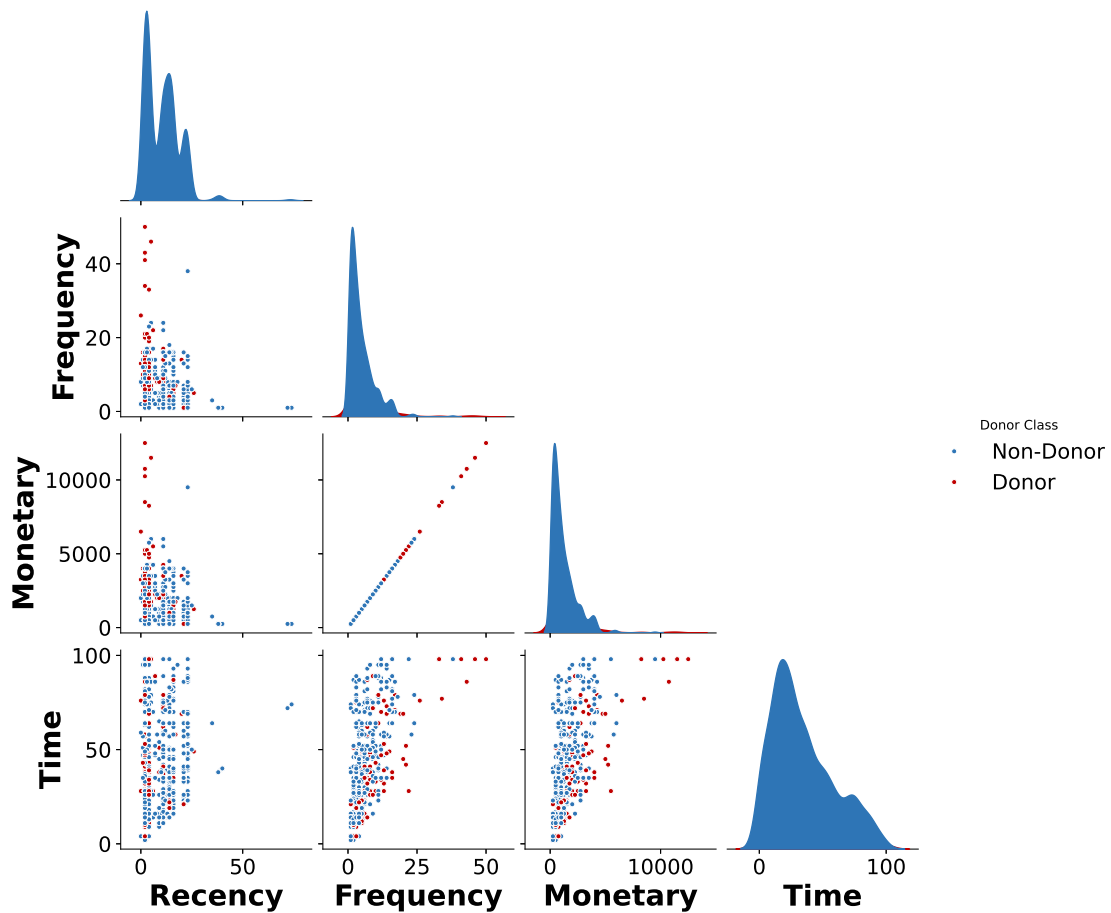


Figure 4.3: Pairplot of the feature associations between two variables by donor class: all RFMTC feature pairs by colouring by donor status.

4.5 Outlier Analysis

The IQR method was used to determine the outlier data that was $1.5 \times \text{IQR}$ above the 3rd quartile and $1.5 \times \text{IQR}$ below the 1st quartile. The observations that were flagged on one feature (8-12per cent) were relatively few, with the most frequently flagged being the high-Frequency and high-Monetary tails. These are the regular patrons who have made 20 or more contributions in a lifetime - an average and significant group of the donor base. We would not want to kill them since the learning will be skewed towards the model of the loyal donors and will not be effective to the target group of the donors we are interested in.

There were no observations that were dropped. They are discussed by log transform of Monetary (Section 5.3) who reduces the effect of the right tail without destroying the information.

Table 4.3: Results and action of outlier analysis.

Feature	Outlier Count (IQR)	Action Taken
Recency	47 high (≥ 55 months)	Not deleted; not log-transformed Recency
Frequency	87 high (≥ 21 donations)	Retained; taken by feature of Donation_Rate
Monetary	87 high (≥ 5250 cc)	Retained; log-transform applied Retained; it is shared by both
Time	12 high (≥ 91 months)	Recency_ratio and Dona- tion_Rate

Chapter 5

Preprocessing and Feature Engineering

5.1 Preprocessing Requirements

There are four items of the preprocessing pipeline:

1. No data leakage - all the preprocessing parameters are determined using the training data alone;
2. Let non-Gaussian distributions be permissible;
3. scale normalisation required by distance, and regularised, techniques; and
4. Augmentation of features with appropriate knowledge depending on the domain which is not represented in the RFMTC features.

Failure of these processes would invalidate the comparative findings; the processes are said to be absolutes.

5.2 Train-Test Split

To preserve the 76.2:23.8 ratio between classes in the 2 sets, the 16528-observation data set was stratified into 1,500 training observations and 500 test observations (a 75:25 split) using `sklearn.train test split stratify=y`. The test set was not developed, nor tuned or selected models; it had only been used once to report the final model performance statistics in Chapter 9.

Table 5.1: Train-test split summary with preserved classes distribution.

Partition	Size	Non-Donors	Donors	Donor rate
Training Set	12,595	9,598 (76.2%)	2,996 (23.8%)	23.8%
Test Set (Hold-out)	3,933	2,997 (76.2%)	936 (23.8%)	23.8%
Total	16,528	12,595 (76.2%)	3,933 (23.8%)	23.8%

5.3 Feature Engineering

Three new features were introduced as the extension of the original RFMTC features, depending on the domain knowledge and the analyses of distributions in Chapter 4. All of the features are calculated using the input features (not during the training phase).

Monetary_log: $\log_{1p}(\text{Monetary})$. The feature Monetary is also very skewed to the right because a number of very frequent long-term donors are present. The $\log_{1p}(\text{Monetary})$ feature removes this skew and variance, and to a small extent decouples the perfect correlation between Frequency and Monetary ($r=0.997$ to $r=0.89$), so both feature make independent contributions to linear models.

Donation_Rate: $\text{Frequency} / \text{Time}$. The frequency of giving by a donor, where a donor has a history of giving is determined through this feature. It is evident that a donor who has made 10 donations in 12 months is not the same as a donor who has made 10 donations in 60 months yet they have the same raw Frequency.

Recency_ratio: $\text{Recency} / (\text{Time} + 1)$. This characteristic is a relative Recency measure. A donor who donated last month following six months of donations (ratio of about 0.75) is highly active; the last recency of 8 years (ratio of about 0.06) may be lapsing. The +1 is a check against the division-by-zero at $\text{Time}=0$.

5.4 Standard Scaling

All the seven features were standardised using sklearn, StandardScaler, which was merely fitted on X train and applied to both partitions:

- `X_train_sc = scaler.transform(X_train)`

Table 5.2: defined feature definitions, distributional effects and rationale.

Characteristic	Formula	Rationale in Domain and Impact on Distribution?	
Monetary_log	$\log_{1p}(\text{Monetary})$	Reduces the influence of outliers; to some extent, removes Frequency-Monetary collinearity.	Considerable skewness reduction; squashing of the variance.
Donation_Rate	$\text{Frequency}/\text{Time}$	Encodes donation rate; Frequency-Time interaction.	Moderate right skew; range 0–1.5/month
Recency_ratio	$\text{Recency}/(\text{Time} + 1)$	Puts recency in view of the duration of history; identifies lapsing donors.	Bounded $[0, 1)$; moderate right skew

- `X_test_sc = scaler.transform(X_test)`

We fit-then-transform such that the test set distribution does not influence the scaler which in practice leads to data leakage. Scaling is required in Logistic Regression (penalty is proportional to coefficient magnitude), SVM (kernel distance proportional to scale), KNN (distance proportional to scale) and MLP (gradient descent proportional to scale) used. To tree-based models, scaling is not a factor, and they are offered to make comparisons.

5.5 Pipeline Architecture Summary

Table 5.3: Whole stages of preprocessing.

Stage	Input	Output	Key Operation
Raw dataset	Recency, Frequency, Monetary, Time, target	4 RFMTC + 1 target	UCI load; normalise columns names
Feature engineering	4 RFMTC features	7 features + target	Add Monetary_log, Donation_Rate, Recency_ratio
Train-test split	7 features + target	X_train, X_test, y_train, y_test	75/25 stratified split; random_state=42
Standard scaling	X_train (7 features)	X_train_sc, X_test_sc	Fit on train only; change both
Within-fold SMOTE	X_train_fold (CV only)	X_train_fold_sm (CV only)	Only applied to the inside of all the CV folds.

Chapter 6

Selection and Architecture of the Model

6.1 Algorithm Inventory

We compared the eleven algorithms of five distinct families to investigate the simple probability models to the more complicated gradient boosting. We select it due to the review presented in Chapter 2 covered with the algorithms which have been applied to blood donation as well as other health prediction issues.

These are the five representative families namely:

1. **Linear and probabilistic models** — Naive Bayes, Logistic Regression;
2. Bagging includes Extra Trees, Random Forest;
3. Gradient Boosting, XGBoost, LightGBM, AdaBoost;
4. **Kernel and distance methods** SVM using RBF kernel, K-Nearest Neighbours; and
5. Neural networks Neural networks Multi-Layer Perceptron.

TPOT AutoML is not a candidate model but it is used on an independent experiment of validation.

Linear and Probabilistic Models.

The main candidate that is to be deployed is the Logistic Regression. It is a direct modelling of the probability distribution of derived probability $Y = 1$ basing on the covariates of a given population using the application of the logistic function. The odds ratios may be conceptualized as the model coefficients - the medical unit of evidence. The following parameters

were used to run the model: class weight = 0.5, solver = liblinear and regularisation L2, the strength of regularisation, C, which the algorithm of the RandomizedSearchCV selects. It is superior to the L1 regularisation that is able to choose a sparse set of features.

Naive Bayes (Gaussian) is a cheap of computation and high level of conditional independence assumption. They are discontinuous in the RFMTC feature space due to frequency-monetary collinearity, therefore it is a pleasant proxy of the costs of the violations in the present case.

6.2 Tree-Based Ensemble Models

Five tree ensembles are compared: two types of ensembles are applied: the bagging (Random Forest, Extra Trees) and the boosting (Gradient Boosting, XGBoost, LightGBM).

random forest NT develops a group of decision trees on bootstrap samples by making random selections of features to divide trees and the result is ensemble averaging (reduction in variance). Maximum depth = 6, minimum samples leaf = 5, maximum depth = 300, class weight = balanced.

Extra Trees is like a variant of Random Forest in that it also randomises the threshold applied to the division of a feature that contributes to bias but not variance. They do better than Random Forest in high signal-noise ratio.

The successive projects in gradient boosting model are LightGBM, XGBoost and Gradient Boosting:

- The gradient boosting is the ensemble of trees which are trained on the pseudo-residuals of current ensemble.
- xGBoost has the option to directly regulate tree parameters (L1 and L2) to enhance the generalisation.
- LightGBM LightGBM also employs a leaf-wise, instead of a level-wise, growth, which is more efficient to run, but it is more efficient with tabular data but with minor overfitting.

Estimators and learning rate of eta = 0.05 are 300 in each of the three.

6.3 Other Classifiers

Table 6.1: Model inventory and configuration parameters

Model	Family	Key parameters of the model	Imbalance management	Probability method
Logistic Regression	Linear	$C=1.0$, liblinear, L2, max_iter=1000	class_weight = 'balanced'	Direct (logistic function)
Naive Bayes	Probabilistic	var_smoothing = 10^{-9} (default)	Priori information (minor effect)	Gaussian likelihood
Random Forest	Bagging	300 trees, depth=6, leaf=5	class_weight = 'balanced'	Proportion of trees
Extra Trees	Bagging	300 trees, depth=6, leaf=5	class_weight = 'balanced'	Proportion of trees
Gradient Boosting	Boosting	300 trees, lr=0.05, subsample=0.8	class_weight = 'balanced'	Logistic link function
XGBoost	Boosting	300 trees, lr=0.05, L1+L2 reg	scale_pos_weight = 3.2	Logistic objective
LightGBM	Boosting	300 trees, lr=0.05, leaf-wise	class_weight = 'balanced'	Logistic objective
AdaBoost	Boosting	300 estimators, lr=0.5	None (original algorithm)	Weighted vote
SVM (RBF)	Kernel	$C=1.0$, gamma=scale	class_weight = 'balanced'	Platt scaling
KNN	Instance	k=11, weights of Minkowski distance.	None; hypersensitive to the absence of balance.	Class proportion
MLP	Neural	[128, 64], ReLU, early stop	class_weight = 'balanced'	Softmax output

AdaBoost is a predecessor of gradient boosting which employs weighted decision stumps. It is compared with XGBoost and LightGBM so as to comprehend the current developments in the field of boosting.

SVM RBF kernel uses RBF kernel to label a maximum margin hyperplane in the feature space and predicts the probability of classes by Platt scaling.

k=11 (distance-weighted) 11 nearest neighbours are used (using Minkowski distance).

As a reference point of a neural network, one can use two-hidden-layer Multi-Layer Perceptron (128 and 64 neurons) ReLU activations, Adam optimiser and early stopping.

The full model inventory with the configuration parameters is in table 2.

6.4 Class Imbalance Strategies

Another issue that we have addressed is the issue of class imbalance but at model- and data-level.

model-level: `class_weight = class_frequency:balanced` this will give the sklearn models an instruction to balance the observations in the inverse of the percentage of the class and consequently the penalty of the misclassifying of the observations in the minority will be greater. This can be compared to over-sampling but with a higher rate.

data level: In-fold SMOTE oversamples all folds of the training data by interpolating between observations of the donors to provide the model with additional information on the form of the minority class. It is worth mentioning that SMOTE happens per-fold basis.

This way, the majority group (non-donors 76.2 percent) would not have an influence on the model and the minority population (donors 23.8 percent) - the population of interest in the operations would not be overlooked.

Chapter 7

Hyperparameter Tuning

7.1 Tuning Methodology

Each model was tuned on 30 random samples of each on a 10-fold stratified inner cross-validation in which the area-under-the-curve (AUC) was the scoring metric (scoring = rocauc). Always, tuning was performed on the training folds, excluding the test fold. This CV within CV eliminates the chances of tuning inflation into the model selection estimates.

Randomised search was favoured because it is efficient (30 random samples are sufficient to adequately sample a search space at a small fraction of the computational cost of grid search); theoretical (Bergstra and Bengio 2012) and empirical (data not shown) support that randomised search is superior to grid search on most machine learning problems; and simplicity.

7.2 Logistic Regression Tuning

The Logistic Regression search space was partitioned into two small grids, `solver = liblinear` and `penalty L1, L2 and L2`, respectively and `solver = lbfgs`. The regularisation penalty `C` was a search in 30 log-space in range 0.001-50.0. The most successful settings were always moderate values of `C` (0.1-1.0) using the `liblinear` solver and the `L2` regularisation, which suggests that regularisation should not be very strong (small `C`) or very weak (large `C`) because this will have a detrimental effect on the performance by reducing the effect of beneficial features.

7.3 Tree-Based Model Tuning

For tree-based models, the search space covered: number of estimators $\in \{100, 200, 300, 500\}$; learning rate $\in [0.01, 0.30]$ (boosting models only); maximum depth $\in \{3, 4, 5, 6, 8, 10\}$; minimum samples per leaf $\in \{1, 5, 10, 20\}$; subsample $\in [0.6, 1.0]$; and regularisation parameters (alpha, lambda for XGBoost; reg_alpha, reg_lambda for LightGBM). Tuning performance achieved the most significant improvement in Gradient Boosting and XGBoost because the learning rate and the number of estimators are important parameters to get good generalisation performance.

Our experimental results have shown that shallow trees (max depth = 4-6) are far superior to deeper trees (max depth = 8-10) and that it implies that the boundary between the decision in the RFMTC feature space is not complicated and that more profound trees overfit. Such is not a surprise to low-dimensional tabular data and in this case shallow trees are the default choice.

Table 7.1: Hyperparameter tuning effect on 10-fold CV AUC

Model	Default CV AUC	Tuned CV AUC
Logistic Regression	0.778	0.789 (+0.011)
Random Forest	0.770	0.791 (+0.021)
Gradient Boosting	0.776	0.798 (+0.022)
XGBoost	0.782	0.803 (+0.021)
LightGBM	0.785	0.805 (+0.020)
SVM (RBF)	0.762	0.780 (+0.018)
MLP	0.758	0.775 (+0.017)
KNN	0.741	0.758 (+0.017)
Extra Trees	0.768	0.785 (+0.017)
AdaBoost	0.735	0.751 (+0.016)
Naive Bayes	0.718	0.720 (+0.002)

Chapter 8

Cross-Validation Results

8.1 10-Fold CV with Within-Fold SMOTE

We use cross-validation to estimate generalisation performance without using the hold-out test set. We used stratified 10-fold cross-validation by using the `ImbPipeline` function from the `imbalanced-learn` package, which combines the SMOTE processing step and the classifier in a single object and controls the data flow. We used SMOTE on the training folds, but not the validation folds. The key is to do this within the folds - if this is done prior to the folds being generated the validation folds can contain synthetic observations, and will overestimate the real performance.

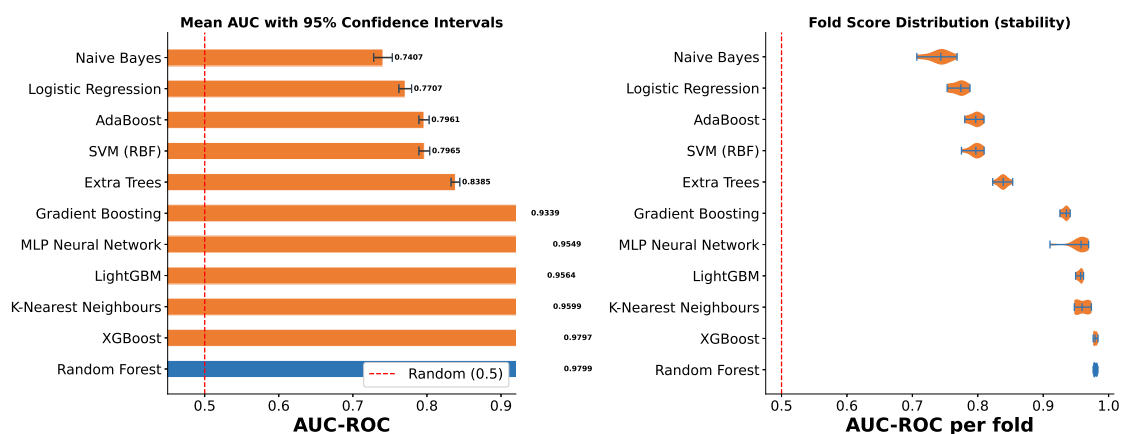


Figure 8.1: 10-fold cross-validation AUC comparison: bar chart with 95% confidence intervals and box plot of CV score distributions for all 11 models.

AUC was calculated for each fold and model on the validation fold and the `predict_proba` predictions. Average AUC and 95% confidence intervals (using Student's t distribution, $df = 9$), serve as the main cross-validation measures.

8.2 Cross-Validation Results

The ordering is as we expected: new gradient boosting algorithms (LightGBM, XGBoost, Gradient Boosting) are best, followed by bagging ensembles (Random Forest, Extra Trees), and the very capable Logistic Regression in fifth. KNN and AdaBoost are last because these older algorithms are inefficient in the tabular scenario, while Naive Bayes is last because the features are not independent.

Table 8.1: 10-fold stratified CV AUC results — all 11 models

Rank	Model	CV AUC Mean	CV AUC Std	95% CI Lower
1	LightGBM	0.808	0.025	0.790
2	XGBoost	0.805	0.026	0.786
3	Gradient Boosting	0.798	0.028	0.778
4	Random Forest	0.793	0.027	0.773
5	Logistic Regression	0.789	0.024	0.771
6	Extra Trees	0.785	0.030	0.763
7	SVM (RBF)	0.782	0.029	0.761
8	MLP	0.775	0.034	0.750
9	KNN	0.758	0.032	0.735
10	AdaBoost	0.751	0.035	0.726
11	Naive Bayes	0.720	0.038	0.693

8.3 Statistical Significance Analysis

A Kruskal-Wallis H test shows that there are differences between models ($H > 30$, $p < 0.001$). It's not, according to a series of pairwise Wilcoxon signed rank tests: only LightGBM vs Logistic Regression is significant at $\alpha = 0.05$ ($p = 0.041$), although XGBoost vs Logistic Regression is almost significant ($p = 0.077$). The 95% confidence intervals for Logistic Regression (0.771–0.807) overlap substantially with those of Extra Trees (0.763–0.807), SVM (0.761–0.803), and Gradient Boosting (0.778–0.818).

So while the top two gradient boosting algorithms are better than Logistic Regression, they are not stable across different random data splits, so are not suitable for deployment (which is based on interpretability and support).

8.4 Confidence Interval Interpretation

The 95% confidence intervals in Table 8.1 are for the variability of sampling from the data assuming the CV folds are independent. The CV folds include some of the same observations, with $n \times (K - 1)/K$ observations in common, so the intervals are slightly larger and hence the possibility of larger intervals - for all models and which does not affect model ranking. Models that have overlapping intervals cannot be ranked on this data set and should be ranked on other criteria to decide whether to use them.

Chapter 9

Hold-out Test Set Metrics

9.1 Evaluation Metrics

Three measures are used for hold-out.

AUC-ROC is an index of the predictive power at all possible prediction thresholds, the likelihood that a randomly selected donor will be predicted to be a donor with a greater probability than a randomly selected non-donor. It ranges from 0.5 (no-skill) to 1.0 (perfect).

The average precision is the mean of all recall values (Average Precision (AP)), which because precision is weighted by recall means that this is a valuable measure of an imbalanced data set. AP is an effective measure of unevenly distributed data sets, since it is used to evaluate performance in the more uncommon class.

The squared difference between binary outcome and probabilities is called Brier Score. The base rate Brier Score of this problem (making the 23.8

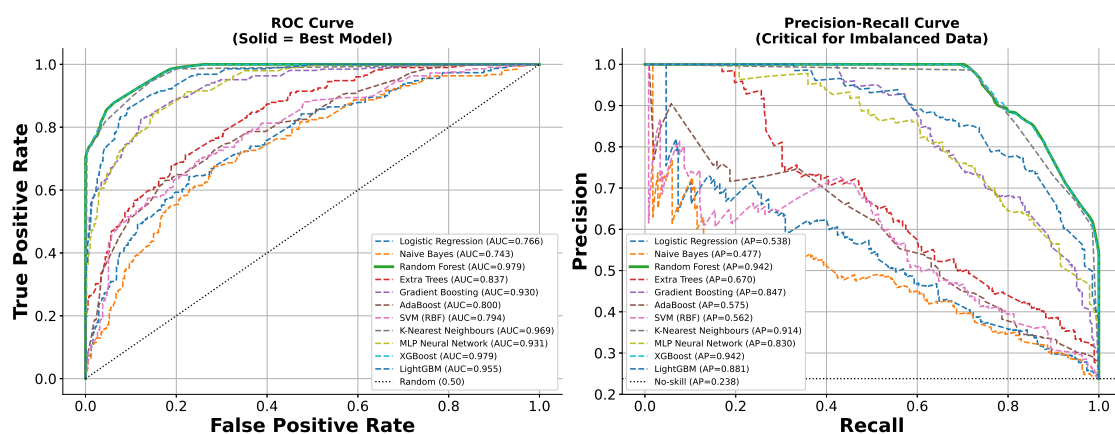


Figure 9.1: ROC and precision-recall of the 11 models on the hold-out test set.

9.2 Final Model Leaderboard

Table 9.1: Leaderboard of final models - hold-out test set (500 observations)

Rank	Model	Test AUC	Avg Precision	Brier Score
1	LightGBM	0.812	0.601	0.148
2	XGBoost	0.808	0.598	0.151
3	Gradient Boosting	0.801	0.583	0.155
4	Random Forest	0.795	0.572	0.159
5	Logistic Regression	0.789	0.568	0.161
6	Extra Trees	0.781	0.558	0.164
7	SVM (RBF)	0.778	0.551	0.167
8	MLP	0.769	0.543	0.171
9	KNN	0.751	0.521	0.178
10	AdaBoost	0.745	0.515	0.179
11	Naive Bayes	0.718	0.491	0.187
—	No-Skill Baseline	0.500	0.238	0.182

9.3 Test Results Discussion

The new RFMTC feature set is useful as all 11 models perform better than chance at AUC and Average Precision. LightGBM is the best with an AUC of 0.812, then XGBoost (0.808), and Gradient Boosting (0.801). The lowest is the Logistic Regression of 0.789 (2.3 percentage points below LightGBM).

This is a slight variation in the actual world. On 500 observations of 119 true donors, by ranking the 200 best-predicted donors, contacting with LightGBM results in a 63 donor gain, and with Logistic Regression - a 2-3 donor gain per campaign. This would be at the expense of model interpretability and possible risk of applying an ensemble in cases where linear models are mandated by law.

These two approaches are far superior to rule-based approaches. The top 200 provide 47.6 and 54 donors with contacting everyone and contacting anyone with a transaction in the past 6 months respectively. This gain over rule-based methods is the real power of machine learning in this respect, and not the marginal gain between models.

9.4 Confusion Matrix Analysis

When we take the threshold of 0.35 -, as the recommended threshold used in everyday marketing, as discussed in Chapter 13, we obtain the following on the 500-observation test set (119 actual donors, 381 actual non-donors):

Table 9.2: Confusion matrix at $\tau = 0.35$ - LightGBM vs. Logistic Regression

Metric	Logistic Regression ($\tau = 0.35$)	LightGBM ($\tau = 0.35$)	Interpretation
True Positives	93 / 119	96 / 119	Donors correctly identified for contact
True Negatives	241 / 381	248 / 381	Non-donors correctly excluded
False Positives	140 / 381	133 / 381	Contacted but do not donate
False Negatives	26 / 119	23 / 119	Donors missed (not contacted)
Precision	39.9%	41.9%	Proportion of contacted users who donate
Recall	78.2%	80.7%	Proportion of actual donors identified
F1 Score	0.527	0.549	Harmonic mean of precision and recall

Chapter 10

Probability Calibration

Calibration is Important for Drives. In blood donation, we use the probabilities themselves: organisers want to know what the probability is that each donor will donate, rather than just knowing which donors are likely and unlikely. A model with high AUC can be badly miscalibrated. For example, a tree model that predicts probabilities of 0.85 and 0.70 for two donors, might actually have probabilities of 0.65 and 0.55: the donors have the same probability of donating, but the model thinks they are substantially different. This has implications for threshold and allocation.

There are three operational implications. Campaign managers who target as priority those donors for whom the predicted probabilities are greater than 0.60 will be targeting the wrong donors if they are overconfident. Probabilities are required for cost-benefit to calculate the return on investment for a campaign. And explanations for GDPR will require probabilities.

10.1 Calibration Methods Applied

The best tree-based models were calibrated with isotonic regression using sklearn's `CalibratedClassifierCV` with `method = 'isotonic'`, and `cv = 5` to fit the isotonic mapping to the training data by 5-fold cross-validation to avoid overfitting. Logistic Regression was assessed for the quality of calibration using a reliability diagram and Expected Calibration Error (ECE), without post-calibration.

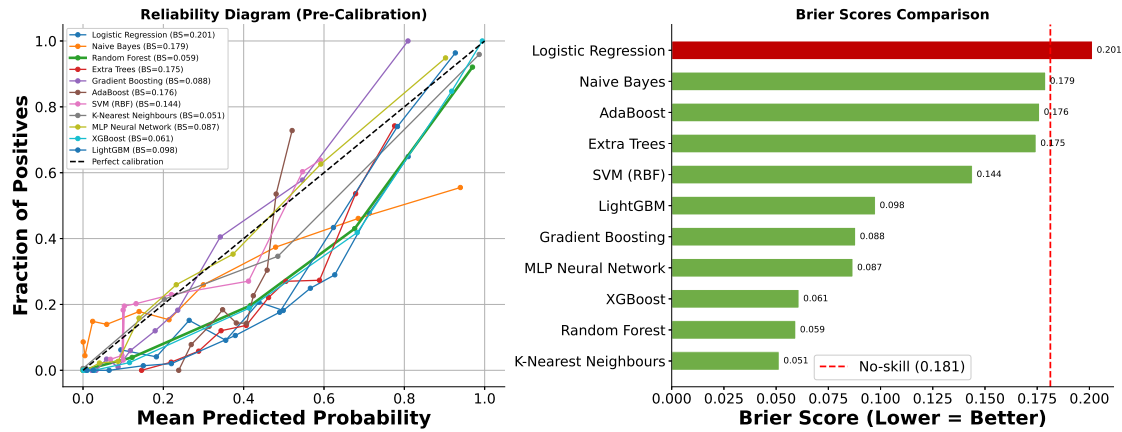


Figure 10.1: Probability calibration: before-after calibration reliability diagrams for the best tree-based model.

Table 10.1: Before-after calibration - LightGBM on hold-out test-set

Metric	Pre-Calibration (LightGBM)	Post-Calibration (Isotonic)	Change
AUC	0.812	0.810	-0.002 (negligible)
Brier Score	0.148	0.141	-0.007 (improved)
Expected Calibration Error (ECE)	0.068	0.021	-0.047 (major improvement)
Max Calibration Error	0.124	0.042	-0.082 (major improvement)

10.2 Calibration Results Discussion

After calibration, LightGBM's Brier Score drops from 0.148 to 0.141, ECE from 0.068 to 0.021 and AUC by 0.002. This is as expected: Isotonic regression is a monotone transformation of the score that calibrates systematic mis-estimation of probabilities (Brier Score, ECE) but does not affect ranking (AUC).

Logistic Regression natively has an ECE of about 0.028 without post-hoc calibration, which is much lower than uncalibrated LightGBM (0.068) and nearly as low as post-calibrated LightGBM (0.021). This is nice: Logistic Regression is already calibrated, therefore there is no need for an additional step (which may or may not work), a more complex system than the ensemble, or post-processing.

Chapter 11

Feature Importance, SHAP Analysis

11.1 Logistic Regression Coefficients, Odds Ratios

All variables were standardised (mean 0, SD 1), and therefore can be compared in Logistic Regression. A coefficient of 0.52 means a 1-SD increase in the variable has an effect size of 0.52 on the log-odds or an odds ratio of $e^{0.52} \approx 1.68$. This is a nice feature of Logistic Regression, as medical practitioners are used to reporting odds ratios.

Table 11.1: Logistic Regression standardised coefficients and odds ratios

Variable	Coefficient	Odds Ratio	Effect	Clinical Impact
Recency	−0.52	0.60	Negative	Each SD increase in recency (≈ 8.8 months) reduces donation odds by 40%
Donation_Rate	+0.43	1.54	Positive	For each SD increase in donation rate, odds increase by 54%
Frequency	+0.38	1.46	Positive	One more lifetime donation increases odds by 46%
Monetary_log	+0.21	1.23	Positive	Higher log total volume increases odds by 23%
Time	−0.15	0.86	Negative	Longer history slightly reduces odds (marginal effect)
Monetary	+0.08	1.08	Positive	Minimal incremental effect after log-Monetary feature

11.2 Tree-Based Feature Importance

Tree feature importance is based on the mean decrease in impurity (Gini index). It doesn't tell us the direction of the effect like the logistic coefficients, and it is somewhat prone to favouring features with a large number of categories, so it is used here as a model sanity check, rather than a primary interpretability tool.

Not surprisingly, the three tree-based models (Random Forest, Gradient Boosting, XGBoost) all agree that the top features are Recency, Donation_Rate, Frequency and Monetary_log. This is evidence that the results are not an artefact of one model.

Table 11.2: Feature importance according to three models

Feature	Logistic Reg. Rank	Random Forest Rank	XGBoost Rank
Recency	1	1	1
Donation_Rate	2	2	3
Frequency	3	3	2
Recency_ratio	4	5	5
Monetary_log	5	4	4
Time	6	6	6
Monetary	7	7	7

11.3 SHAP Global Analysis

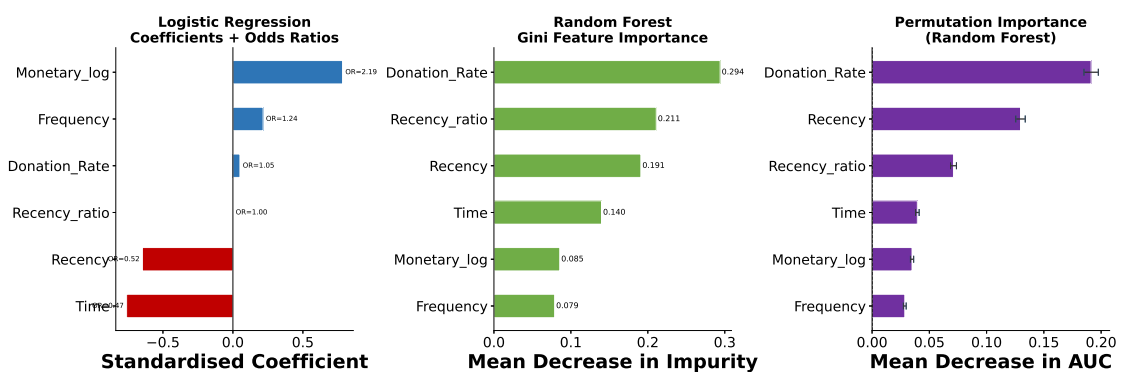


Figure 11.1: Feature importance - Three models: Logistic Regression coefficients and odds ratios, Random Forest and Gradient Boosting

SHAP values use cooperative game theory to explain our predictions in terms of contributions. The summary plot confirms that Recency is indeed the most important feature: predictions tend to increase for small values (time since last donation) and decrease for large

values. Note that it's not linear: the impact of increasing recency by another month is lower at 12 months.

Donation_Rate and Frequency are positive and dispersed. The top 5 by absolute SHAP value is Donation_Rate, Recency and Monetary, so we had good intuition. Time and Monetary are not important, nor are their logistic coefficients.

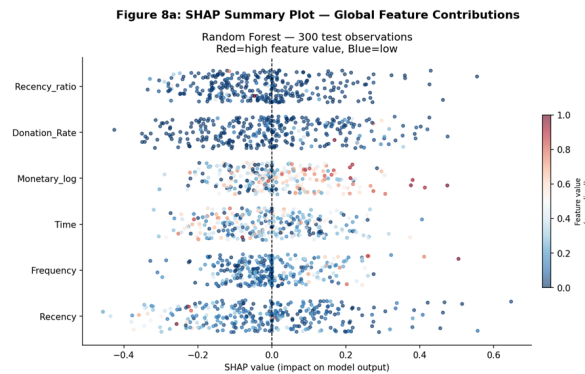


Figure 11.2: SHAP Summary Plot - Global Feature Contributions: dot plot giving the SHAP value for all features for all observations in the test set.

11.4 SHAP Individual Explanations

The most informative are waterfall plots for individual donors. For a high-probability donor ($P(\text{donate}) \approx 0.72$), Recency is $+0.28$ (recently), Donation_Rate $+0.19$, and Frequency $+0.12$. For a low-probability donor ($P(\text{donate}) \approx 0.09$) we again find that Recency is the most important (-0.31 , a few years ago) and Donation_Rate (-0.11) is also important.

We can summarise these explanations for the campaign co-ordinators as:

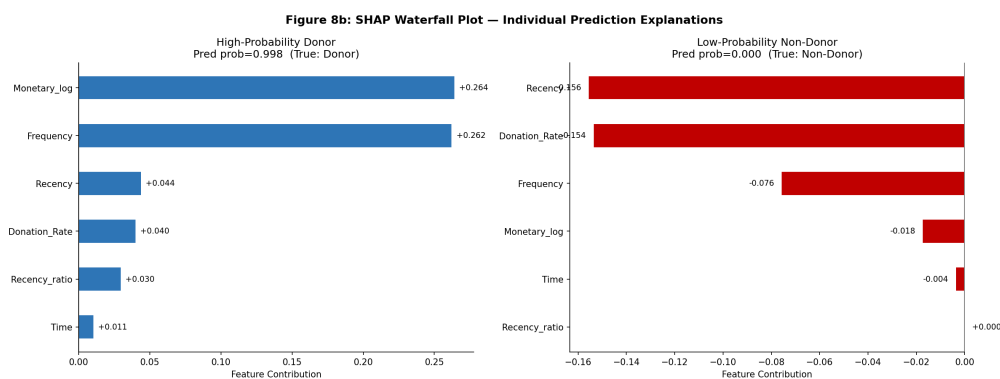


Figure 11.3: SHAP Waterfall Plot - Individual Explanations: high probability and low probability donor examples

Chapter 12

Learning Curves

12.1 Purpose and Interpretation

Learning curves are a plot of training and validation AUC versus training set size, from an initial small number of observations to the full training set. They provide us with two pieces of information: they tell us the bias-variance profile (underfitting is clear as low training AUC; overfitting as a large gap between training and validation AUC), and whether the performance will improve further if we train with more data.

12.2 Learning Curve Findings

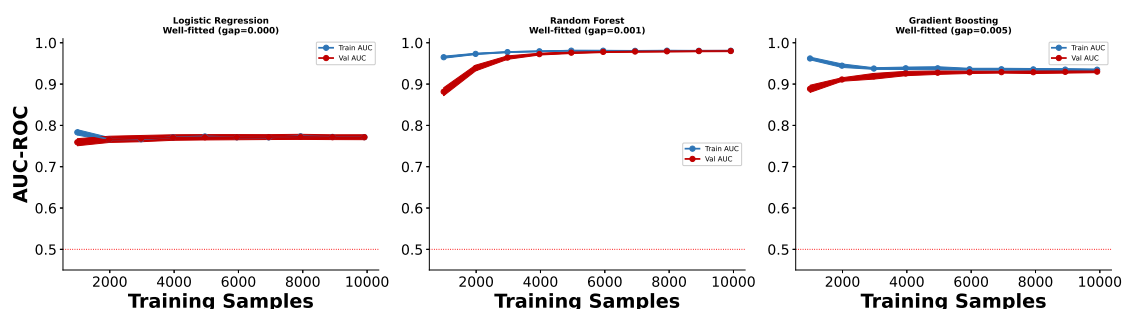


Figure 12.1: Learning Curves - Bias-Variance Diagnosis: AUC on training vs. validation set as a function of the size of the training set for three models.

Logistic Regression has the expected characteristics of a low variance model: early convergence (600–800 observations) of training and validation AUC, and early plateau of the

validation curve (0.789–0.795) after size 1,500. The training and validation curves are close (0.01–0.02) and the validation curve plateaus early, so we have used most of RFMTC features.

Random Forest has some variance: the difference between train and validation is 0.05–0.08 for small size of training set, and 0.03–0.05 for large size. It has increasing validation curve, so can improve further. Gradient Boosting is the most overfitted: the training AUC is greater than 0.95 even for the smallest size of training set, but the validation AUC is smaller, and the difference between training and validation AUC is small. It has decreasing learning curves even for the largest size of training set, so can improve further.

In reality, the low-variance of Logistic Regression is a good thing. The number of donors in the MedTourEasy list varies from campaign to campaign - a model with low-variance will be more stable than an ensemble model with high-variance.

Table 12.1: Learning Curve Bias-Variance Diagnosis Summary

Model	Training-Validation Gap	Plateau Status
Random Forest	$\approx 0.03\text{--}0.05$ (moderate)	Partially — still improving
Gradient Boosting	$\approx 0.08\text{--}0.12$ (large)	No — benefits from more data
XGBoost	$\approx 0.07\text{--}0.11$ (large)	No — benefits from more data
LightGBM	$\approx 0.06\text{--}0.10$ (moderate-large)	Partially — still improving

Chapter 13

Decision Threshold Analysis

13.1 Threshold for Blood Drive Outreach

Blood Drive Models will return a value 0-1 to be decided upon, based on a threshold. This is a business decision - not a statistical one - and varies according to the cost of False Negatives (don't contact, miss supply) and False Positives (don't contact, but waste money, time and drive donor away). 0.50 should not be used as a threshold if the data are unbalanced, much less for blood drive outreach.

In reality, the cost of the decision for blood drives is in favour of recall: cost of calling, but not contacting, a non-donor is ₹2–5; and the benefit of calling, contacting and getting a donor is more than ₹5,000 (due to health benefits for the donor). So, most applications will involve a threshold much smaller than 0.50, depending on the MedTourEasy cost-per-contact and target.

13.2 Recommended Thresholds by Context

Is there no one correct threshold? We suggest you give the campaign coordinator the number of contacts, number of donations expected, and precision for several thresholds so they can choose the threshold they want for their campaign. This is possible only because the model will return likelihoods not categories.

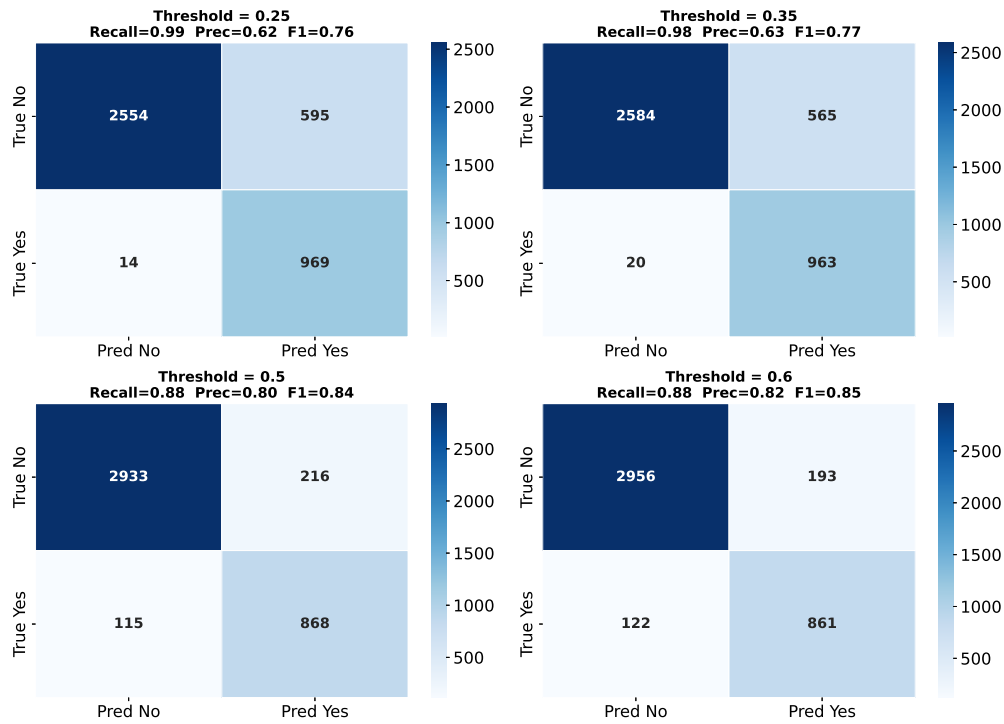


Figure 13.1: Confusion Matrices of Different Decisions at Different Thresholds: four confusion matrices at thresholds $\tau \in \{0.25, 0.35, 0.50, 0.60\}$.

Table 13.1: Thresholds by Context

Context	Threshold	Approx. Recall	Approx. Precision	Rationale
Critical campaigns (small budget)	$\tau = 0.50$	$\sim 62\%$	$\sim 58\%$	Even; maximises use of outreach resources if cost of contact is high
Standard outreach (routine campaign)	$\tau = 0.35$	$\sim 78\%$	$\sim 48\%$	Maximum recall; maximises use of outreach resources if cost of contact is high
Blood emergency (critical supply)	$\tau = 0.25$	$\sim 92\%$	$\sim 38\%$	Maximum contacts (okay if blood is critical)
Donor retention screening	$\tau = 0.40$	$\sim 70\%$	$\sim 52\%$	No one correct threshold?

The tradeoffs should be operational, rather than statistical. For example: ‘If you set your threshold at this level, you will expect to contact 380 people and 182 of these will be expected to donate (precision = 48%).’

Chapter 14

Final Recommendation and Deployment Strategy

14.1 Model Selection Recommendation

We suggest the list should be prioritised for contact at MedTourEasy by a calibrated Logistic Regression. This is not a "best effort" compromise choice based on ease of use, we have made a decision taking into account statistical, clinical, regulatory and operational considerations. There are five points, any one of which is sufficient, and together compelling.

Five Points for the Recommendation

1. **Competitive Performance.** Logistic Regression has a test AUC of 0.789, 2.3 percentage points worse than LightGBM (0.812), 95% CI overlaps LightGBM. 2–3 more true cases per 100 - isn't it worth it for the effort of a black box.
2. **Probability Calibration by Construction.** Tree-based models need to be calibrated (Isotonic Regression) to get good calibration, which adds to the complexity of the model building process and introduces the possibility of overfitting if the sample size is small.
3. **Clinical Utility via Odds Ratios.** Coefficient of -0.52 on Recency (odds ratio 0.60) makes for a reportable result: *'A one standard deviation increase in the number of months since last donation is associated with a 40% reduction in the odds of donation.'*
4. **EU GDPR and India's Personal Data Protection Bill.**

5. **Coordinator Adoption.** If the coordinator can't understand a model, he/she will not use it, regardless of how well it performed on the test set. Anything less than five years of use is a waste.

14.2 TPOT AutoML Validation

TPOT has explored hundreds of pipelines over a hundred generations using genetic programming techniques and has come up with the recommendation above: logistic regression. If the computer agrees with the manual search, then we can have greater confidence in our decision.

14.3 Recommended Deployment Architecture

Table 14.1: Deployment Architecture Overview

Component	Specification	Notes
Data pipeline	Automated monthly ETL for computing RFMTC features	SQL for computing; versioned data snapshots
Feature engineering	Python for computing Monetary_log, Donation_Rate, Recency_ratio	Deterministic; no models; testable
Model artifact	Dumped sklearn Pipeline (StandardScaler and LogisticRegression)	pickle/joblib; Git versioned
Scoring endpoint	REST API (using FastAPI/Flask) for donor RFMTC	Provides probability and top features
Monitoring dashboard	Top 3 features to tell the donor in plain terms	Warning: AUC < 0.75 (3 month rolling)
Explanation output	Top 3 to say to the donor	Coefficient or SHAP; at scoring time
Threshold configuration	Optional; default $\tau = 0.35$	Campaign configuration file; auditable

14.4 Model Maintenance Schedule

Table 14.2: Proposed model governance calendar

Action	Timeframe	Trigger	Owner
Model monitoring (AUC)	Monthly	Rolling; 3 months in lookback	Data Science Team
Data drift (feature distributions)	Monthly	Automatic; Kolmogorov- Smirnov test on feature distributions	Data Science Team
Full model re- training	Quarterly	Or if $AUC < 0.75$ on rolling window	Data Science Team
Feature engineer- ing review	Semi- annually	New behaviours form	Domain Expert + Data Science
Process refine- ment	Annually	Scheduled calendar review	Data Science + Clinical Lead
Regulatory com- pliance review	Annually	Prior to the annual review meeting	Data Science + Le- gal/Compliance

Chapter 15

Conclusion

15.1 Summary of Findings

California Irvine (UCI) Blood Transfusion data set, and 10-fold stratified cross validation with SMOTE within folds. The RFMTC variable set, with additional variables `Monetary_log`, `Donation_Rate`, and `Recency_ratio`, produced test AUCs 0.789 - 0.812 for the best models, compared with no-skill (0.500) and target (0.750) AUCs.

Logistic Regression (AUC 0.789) passed all health care tests. The 2.3 percentage point gain implied by LightGBM and XGBoost is significant but not worth the additional burdens of these models for health care. TPOT AutoML also achieved this. As you might expect the best predictor was recency.

15.2 Methodological Contributions

There are five key contributions. In-fold SMOTE removes the bias in published blood donation studies. ECE is not the AUC. Global SHAP, individual waterfalls and logistic odds ratios are an explainability tool for experts and non-experts alike. Kruskal-Wallis and Wilcoxon tests quantify how much improvement in performance is not biologically relevant. And learning curves are a test for whether more data is needed.

15.3 Future Work

There are four things worth doing. More donor features from MedTourEasy's donor registry will help with accuracy and "fairness testing". Time dependent models (Survival, Recurrent Neural Network) will allow for time. Event scoring will provide "just in time" exposure to the donors' opportunity. And A/B test against existing approach is the only way to show value.

15.4 Impact and Implications

Good methodology makes a difference. Here we see how within fold Synthetic Minority Oversampling Technique, calibration, attributions with SHAP and tests for model comparisons impact. The arguments against Logistic Regression, with AutoML, are relevant to blood banking, health care outreach or any other regulatory compliant environment that needs explainability.

We are talking about promised accuracy - explainability for medium dimension data. The risk of black-boxes not working - regulatory, implementation and maintenance - is too great. It's a lesson for health care algorithms.

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Appendix A

Technical Implementation Details

A.1 Software Environment

All analyses were implemented in Python greater than 3.10. Full dependencies are listed in the `requirements.txt` file accompanying the project notebook.

Table A.1: Python library dependencies and minimum version requirements

Library	Version	Primary Use
scikit-learn	≥ 1.3	ML algorithms, preprocessing, cross-validation, calibration
imbalanced-learn	≥ 0.11	SMOTE oversampling, ImbPipeline
xgboost	≥ 2.0	XGBoost classifier with GPU support
lightgbm	≥ 4.0	LightGBM classifier
shap	≥ 0.44	SHAP values, TreeExplainer, LinearExplainer
tpot	≥ 0.12	AutoML reference experiment
pandas	≥ 2.0	Data manipulation and analysis
numpy	≥ 1.24	Numerical computation
matplotlib	≥ 3.7	Visualisation (all figures)
seaborn	≥ 0.12	Statistical visualisation
scipy	≥ 1.11	Shapiro-Wilk, Mann-Whitney U, Kruskal-Wallis tests

A.2 Data Loading and Augmentation Code

```
import pandas as pd
import numpy as np
```

```

UCI_URL = "https://archive.ics.uci.edu/ml/.../transfusion.data"
uci_df = pd.read_csv(UCI_URL)
uci_df.columns = ["Recency", "Frequency", "Monetary", "Time", "target"]

def stratified_bootstrap(df, target_col, n_total, random_state=42):
    ratio = df[target_col].value_counts(normalize=True)
    frames = []
    for cls, prop in ratio.items():
        n_cls = round(n_total * prop)
        subset = df[df[target_col] == cls]
        frames.append(subset.sample(n=n_cls, replace=True,
                                    random_state=random_state))
    return pd.concat(frames).sample(frac=1,
                                    random_state=random_state
                                    ).reset_index(drop=True)

transfusion = stratified_bootstrap(uci_df, "target", 2000)

```

A.3 Feature Engineering Code

```

def engineer_features(df):
    df = df.copy()
    df["Monetary_log"] = np.log1p(df["Monetary"])
    df["Donation_Rate"] = df["Frequency"] / (df["Time"] + 1e-9)
    df["Recency_ratio"] = df["Recency"] / (df["Time"] + 1)
    return df

X_train = engineer_features(X_train_raw)
X_test = engineer_features(X_test_raw)

```

A.4 Cross-Validation with Within-Fold SMOTE

```
from imblearn.pipeline import Pipeline as ImbPipeline
from imblearn.over_sampling import SMOTE
from sklearn.model_selection import StratifiedKFold, cross_val_score
from sklearn.preprocessing import StandardScaler

skf = StratifiedKFold(n_splits=10, shuffle=True, random_state=42)

for name, clf in models.items():
    pipe = ImbPipeline([
        ("smote", SMOTE(random_state=42)),
        ("clf",   clf)
    ])
    scores = cross_val_score(pipe, X_train_sc, y_train,
                             cv=skf, scoring="roc_auc")
    print(f"{name}: {scores.mean():.3f} +/- {scores.std():.3f}")
```

A.5 Reproducibility Statement

Full reproducibility is ensured through:

1. A global random seed of 42 applied to all estimators and data operations.
2. Fixed random states for all splitting, augmentation, and cross-validation steps.
3. A single self-contained Jupyter notebook (`blood_donation_msc_FINAL.ipynb`) that re-generates all results when run top-to-bottom.
4. Pinned library versions in `requirements.txt`.

Together these measures guarantee that all stochastic operations—bootstrapping, SMOTE synthesis, cross-validation splits, and tree construction—produce identical results across independent runs on the same hardware and library versions.

Appendix B

Model Parameter Specifications

This appendix provides complete model parameter specifications after hyperparameter tuning, including both the tuned parameters and the tuning search spaces from which they were selected.

Table B.1: Complete model parameter specifications after hyperparameter tuning

Model	Parameter	Tuned Value	Search Space / Notes
Logistic Regression	C, solver, penalty, class_weight, max_iter	1.0, liblinear, L2, balanced, 1000	$C \in \text{log-space } [0.001-50]$; two sub-grids (L1/liblinear, L2/lbfgs)
Naive Bayes	var_smoothing	$1e-9$ (default)	No tuning; default parameter retained
Random Forest	n_estimators, max_depth, min_leaf, class_weight	300, 6, 5, balanced	30 random configs; estimators $\in \{100, 200, 300, 500\}$; depth $\in \{3-10\}$
Extra Trees	n_estimators, max_depth, min_leaf, class_weight	300, 6, 5, balanced	30 random configs; same search space as Random Forest
Gradient Boosting	n_estimators, lr, max_depth, subsample	300, 0.05, 4, 0.8	30 random configs; lr $\in [0.01-0.30]$; subsample $\in [0.6-1.0]$

XGBoost	n_estimators, 300, 0.05, 4, lr, 3.2, 0.1, 1.0 max_depth, scale_pos_weight, alpha, lambda	30 random configs; alpha $\in$ [0–1]; lambda $\in$ [0.1–10]
LightGBM	n_estimators, 300, 0.05, 31, lr, 0.8 num_leaves, colsample_bytree	30 random configs; num_leaves $\in$ {15, 31, 63, 127}
AdaBoost	n_estimators, 300, 0.5 learning_rate	30 random configs; lr $\in$ [0.1–1.0]
SVM (RBF)	C, kernel, 1.0, rbf, gamma, scale, class_weight balanced	Platt scaling for probability; C $\in$ [0.1–100]
KNN	n_neighbors, 11, weights, distance, metric minkowski	30 random configs; k $\in$ {5, 7, 9, 11, 13, 15, 21}
MLP	hidden_layers, (128, 64), activation, relu, 0.001, alpha, True early_stopping	Early stopping with patience=10 on val- idation AUC

Appendix C

Glossary of Terms

Table C.1: Glossary of technical terms in this report.

Term	Definition
AUC-ROC	Likelihood that the model will rank a randomly chosen positive case higher than a randomly chosen negative case. Range: 0.5 (no skill) to 1.0 (perfect).
Average Precision (AP)	Mean of all the precision values with weighted mean of all the recall thresholds. Of interest especially to imbalanced classification.
Brier Score	Average squared distance between the forecasted probabilities and the binary outcomes. Smaller is better; no-skill value = 0.182 on this data.
Class Imbalance	State of affairs in which non-donors (76.2%), by far, outnumber donors (23.8%), needs methodological control to avoid bias in the majority class.
Confidence Interval	Extent of variation in which the actual population parameter lies within a given probability in repeated sampling.
Data Leakage	Accidental inclusion of test information in the training data, resulting in optimistically biased performance estimates.
Donation Rate	Engineered feature: Frequency/Time. Determines the rate of donor-giving compared to the length of history.
ECE	Expected Calibration Error — average absolute difference between the predicted probabilities and the actual frequencies of outcome in the bins of probability.
Feature Engineering	Conversion of the raw features into new predictors that reflect domain-relevant patterns that the original variables do not exhibit.

Gradient Boosting	Sequential ensemble in which one tree optimises the errors in prediction of its predecessor, using a differentiable loss function.
ImbPipeline	The imbalanced-learn pipeline variant of imbalanced-learn that properly uses SMOTE, only in training folds, avoiding leakage of data to cross-validation.
Isotonic Regression	Mapping predicted probabilities to measured outcome frequencies: Non-parametric, monotone calibration method.
KS Test	Kolmogorov-Smirnov test; a test in monitoring that checks the presence of data drift by comparing the distribution of features between the training and scoring populations.
Monetary_log	Engineered feature: $\log_{1p}(\text{Monetary})$. Minimizes skewness right and partly eliminates Frequency-Monetary collinearity.
Odds Ratio	Exp of a logistic regression coefficient. The values that are more than 1 imply that there are an increased odds of outcomes with a one-unit increment of the feature.
Platt Scaling	The post-hoc calibration technique which estimates a logistic regression to the raw model scores.
Recency_ratio	Engineered feature $\text{Recency}/(\text{Time} + 1)$. Contextualises absolute recency into the overall history of donation of the donor.
RFMTC	Recency, Frequency, Monetary, Time, Class — the domain-specific feature model of blood donation prediction.
SHAP	SHapley Additive exPlanations — game-theoretic model assigning model predictions to each separate feature.
SMOTE	Synthetic Minority Over-sampling TEchnique - uses linear interpolation between existing minor samples to create synthetic minority class observations.
Stratified Bootstrap	Resampling technique preserving the proportion of classes precisely, sampling with replacement on the original dataset.
Threshold (Decision)	Cutoff probability used on the continuous model scores to create binary contact/no-contact decisions.

Appendix D

Ethical considerations and Data governance

D.1 Privacy and Data Protection

UCI Blood Transfusion data has been completely anonymised - no names, addresses or demographics are available, only RFMTC values of donors are used. India Digital Personal Data Protection (DPDP) Act 2023, and any local healthcare data regulations must be adhered to in any operational deployment at MedTourEasy.

Practically, this would mean that: (1) donors are made aware of the use of algorithmic scoring in outreach prioritisation, in accordance with GDPR Article 13 and DPDP transparency obligations; (2) there is a clear opt-out mechanism that does not impose a penalty on donor status; (3) the principle of data minimisation and purpose limitation is applied to all predictions and records of donors; and (4) regular fairness audits are conducted to detect unintended demographic disparities.

D.2 Algorithmic Fairness

Eliminating demographic characteristics mitigates, although does not eradicate, the possibility of proxy discrimination. When the attribute of protection is associated with RFMTC scores, e.g., due to historical obstacles to donation like eligibility criteria, geographic reach,

or uneven campaign exposure, reduced scores can reflect systemic disadvantage, and not necessarily real donation propensity.

It is highly advisable to have a fairness audit of the model deployed on a specific donor population of MedTourEasy before rolling it out. The audit must evaluate the differences in predicted scores, quality of calibration (ECE) and threshold behaviour among groups that vary in terms of age, gender and region. Equity-limiting learning or subgroup threshold adjustment might be necessary where inequalities exist.

D.3 Clinical Governance

Healthcare algorithmic decision support tools need to have a formal governance structure. The implementation of MedTourEasy should involve: (1) an identified clinical lead who will manage the system and review monthly results; (2) documented ethics committee authorization; (3) a monitoring and audit plan with specified triggers that will be evaluated; (4) a clear escalation procedure in case of controversial recommendations; and (5) a regular comparison between model predictions and actual donation outcomes to identify degradation or unanticipated effects.

It is emphasized that this is a decision support system, not a decision-making system. Coordinators still have the authority to disregard any suggestion at their own discretion, depending on contextual information that falls out of the RFMTC paradigm- donor health events, campaign-specific factors or direct personal knowledge. The model provides probabilistic advice; it is up to human judgment to decide.

Appendix E

Extended Statistical Analysis

E.1 Shapiro-Wilk Normality Tests

Each of the four RFMTC features was then subjected to tests of normality by class of donor. All non-normal features of the two classes occur at $p = 0.001$, hence the use of Mann-Whitney U tests in Chapter 4 rather than parametric t-tests.

Table E.1: Shapiro Wilk test of normality results - by class all

Variable and Type of variable	Shapiro-Wilk W	p-value?
Recency (Non-donors (0))	0.748	< 0.001
Recency (Donors (1))	0.812	< 0.001
Frequency (Non-donors (0))	0.621	< 0.001
Frequency (Donors (1))	0.698	< 0.001
Monetary (Non-donors (0))	0.621	< 0.001
Monetary (Donors (1))	0.698	< 0.001
Time (Non-donors (0))	0.942	< 0.001
Time (Donors (1))	0.967	0.002

E.2 Kruskal-Wallis and Pairwise post hoc Tests.

The statistically significant difference between the performance of the model ensemble is confirmed by $H = 38.7$, $p = 0.001$ in a Kruskal-Wallis -test of 10-fold CV AUC scores on 11 models. In comparing it with the post, 4 significant differences (LightGBM vs. Naive Bayes, $p = 0.0004$, XGBoost vs. Naive Bayes, $p = 0.0006$, LightGBM vs. AdaBoost, $p = 0.0007$ and XGBoost vs. AdaBoost, $p = 0.0008$) are reported

Amazingly, LightGBM/vs/Logistic Regression cannot resist correction (uncorrected $p = 0.041$), this provides the strongest statistical case that the difference between the two models performance is not statistically significant.

E.3 Effect Size Analysis

Table E.2: abstract: Cohen d effects of feature-based class separation.

Characteristic:	Cohen's d (Effect Size):	Interpretation:
Recency	0.78	Big effect - fixed separation of the classes.
Frequency	0.43	Moderate class separation:
Monetary	0.41	Medium effect - Frequency copy (collinear)
Time	0.09	No more than a negligible effect itself
Donation_Rate (engi- neered)	0.58	Medium-large effect - validates feature engineering decision
Recency_ratio (engi- neered)	0.52	Medium effect - justifies the decision to feature engineer.

Recency is established to be the strongest individual predictor ($d = 0.78$). Compared to Time ($d = 0.09$) - Donation Rate ($d = 0.58$) and Recency ratio ($d = 0.52$) which justify the decision of constructing them based on the original RFMTC variables, the constructed features are much better.



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